

Chapter 4 Lab: Using Script Files and Managing Data

Instructions:

- Write a script file to do the exercises below.
- Make sure to write comments in your code and separate output display for clarity.
- Submit your script file in the following format: FirstName_LastInitial_ch4lab.m
Ex: Ryan Cao would submit his lab as: **ryan_c_ch4lab.m**

1.

The Heat Index HI , calculated from the air temperature and relative humidity, is the apparent temperature felt by the body. An equation used by the National Weather Service for calculating the HI is given by:

$$HI = c_1 + c_2T + c_3R + c_4TR + c_5T^2 + c_6R^2 + c_7T^2R + c_8TR^2 + c_9T^2R^2$$

where

- HI = heat index (in degrees Fahrenheit)
- T = ambient [dry-bulb temperature](#) (in degrees Fahrenheit)
- R = relative humidity (percentage value between 0 and 100)

$$\begin{array}{lll} c_1 = -42.379, & c_2 = 2.049\,015\,23, & c_3 = 10.143\,331\,27, \\ c_4 = -0.224\,755\,41, & c_5 = -6.837\,83 \times 10^{-3}, & c_6 = -5.481\,717 \times 10^{-2}, \\ c_7 = 1.228\,74 \times 10^{-3}, & c_8 = 8.5282 \times 10^{-4}, & c_9 = -1.99 \times 10^{-6}. \end{array}$$

where T is the temperature in degrees F and R is the relative humidity in integer percentage. Write a MATLAB program in a script file that calculates HI . For input the program asks the user to enter values for T and R . For output the program displays the message: "The Heat Index temperature is: XX," where XX is the value of the heat index rounded to the nearest integer. Execute the program entering $T = 90^\circ\text{F}$ and $R = 90\%$.

2.

The monthly saving P that has to be deposit in a saving account that pays an annual interest rate of r in order to save a total amount of F in N years can be calculated by the formula:

$$P = \frac{F(r/12)}{(1 + r/12)^{12N} - 1}$$

Calculate the monthly saving that has to be deposit in order to save \$100,000 in 5, 6, 7, 8, 9, and 10 years if the annual interest rate is 4.35%. Display the results in a two-column table where the first column is the number of years and the second column is the monthly deposit.

3.

The monthly payment, P , of a N years mortgage of an amount L that with a yearly interest rate of r is given by:

$$P = L \frac{\frac{r}{100 \cdot 12} \left(1 + \frac{r}{100 \cdot 12}\right)^{12N}}{\left(1 + \frac{r}{100 \cdot 12}\right)^{12N} - 1}$$

where r is in % (e.g., 7.5% entered as 7.5). Write a MATLAB program in a script file that calculates P . When the program is executed it asks the user to enter the mortgage amount, the number of years, and the interest rate. The output is displayed in the following format: “ The monthly payment of a XX years XXXXXX.XX mortgage with interest rate of XX.XX percent is \$XXXXX.XX”, where XXX stands for the corresponding quantities. Use the program for determining the monthly payment of a \$250,000 mortgage for 30 years and 4.5% yearly interest rate.

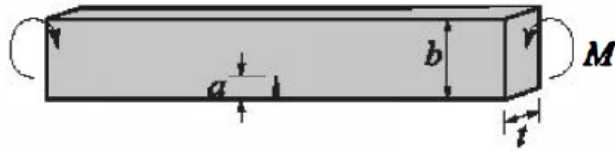
4.

The stress intensity factor K at a crack in a beam exposed to pure bending M is given by:

$$K = C\sigma\sqrt{\pi a}$$

where $\sigma = \frac{6M}{tb^2}$. a is the crack length, b is the width, t is the thickness, and C is a parameter that depends on the geometry of the specimen and crack. For the case of pure bending,

$$C = \sqrt{\frac{\tan\beta}{\beta}} \left[\frac{0.923 + 0.199(1 - \sin\beta)^2}{\cos\beta} \right] \quad \text{where } \alpha = a/b \text{ and } \beta = (\pi\alpha)/2$$



Write a program in a script file that calculates the stress intensity factor K . The program should read the values of M , b , t , and a from an ascii text file using the `load` command. The output should be in the form of a paragraph combining text and numbers,— i.e., something like: “The stress intensity factor for a beam that is 0.25 m wide and 0.01 m thick with an edge crack of 0.05 m and an applied moment of 20 N-m is XX Pa-sqrt(m).” where XX stands for the value of K . Use the program to calculate K when $M = 20$ N-m, $b = 0.25$ m, $t = 0.01$ m, and $a = 0.05$ m.

Output:

Problem 1:

Please enter the temperature in deg F: 90
Please enter the relative humidity in percent: 90

The Heat Index Temperature is: 122

Problem 2:

Years	Monthly Deposit
5.00	1494.99
6.00	1218.02
7.00	1020.55
8.00	872.78
9.00	758.13
10.00	666.67

Problem 3:

Please enter the mortgage amount: 250000
Please enter the number of years: 30
Please enter the interest rate in percent: 4.5

The monthly payment of a 30 years 250000.00 mortgage
with interest rate of 4.50 percent is \$1266.71

Problem 4:

The stress intensity factor for a beam that is 0.25 m wide and 0.01 m thick
with an edge crack of 0.05 m and an applied moment of 20 is 82836 pa-sqrt(m).